

Development and Evaluation of an Open-Source Tool for Machine Scheduling in Forest Operational Planning

by

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The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

Development and Evaluation of an Open-Source Tool for Machine Scheduling in Forest Operational Planning

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Abstract

This thesis describes an open-source modelling framework for forest operational planning. The forest operational planning problem is well researched and tackled from many different perspectives, methods and in many geographic regions and operational environments. However, the answers given to this problem have not been written to be reproducible. Written in Python programming language, the proposed modelling framework maintains open-source availability and accessibility with the goal of amplifying reproducibility and open innovation in forest operational planning. For both the framework and model presented, the design considerations are carefully outlined and explained. Practical applications and implications of the open-source modelling framework are discussed.

Lay Summary

The lay or public summary explains the key goals and contributions of the research/scholarly work in terms that can be understood by the general public. It must not exceed 150 words in length.

Managing forest operations is a complex and important task. Over time, many models have been developed to support forest planning in different locations and situations. These models have improved as data and computing power have advanced. However, one major challenge remains: the results of these models are often difficult to reproduce. To address this, we have developed a new open-source modelling framework in Python for forest operational planning. The main goals of this framework are to keep it openly available and easy to access, which will support better reproducibility and encourage innovation in operational forest planning. This framework uses computational and quantitative methods to test a wide range of planning scenarios, including those that have not been explored before. This helps users adapt more quickly to changes in the forest environment, a needed quality given climate change and increasing complexity of forest management decisions.

Preface

The Preface must contain the following:

- A statement detailing your contribution to the identification and design of the research program, performance of the various parts of the research, and analysis of the research data.
- A list of any publications arising from work presented in the dissertation, and the chapter(s) in which the work is located. There must also be a statement detailing the relative contributions of all collaborators and co-authors (including supervisors and members of the supervisory committee) and stating the proportion of research and writing conducted by the student.

All the work presented in this thesis was carried out by the author, Rosalia Jaffray, during her MASc program. The work was conducted under the supervision of Dr. Gregory Paradis at the Forest Resources and Ecosystem Services Hub of the University of British Columbia, Vancouver, Canada. It was realized in collaboration with Dr. Dominik Roeser and Dr. Ken Byrne, Rosalia's Supervisory Committee members, as well as Dr. Kathleen Coupland.

This document has been prepared using the thesis-by-chapter approach and includes two journal articles developed as part of my graduate studies. My contributions to these papers are as follows: literature review, model development, validation, and writing of the manuscripts.

The co-authors contributed to the papers in the following components: model development, manuscript editing, general support and good vibes.

The research gaps and methods to address those gaps were identified by myself and Dr. Gregory Paradis.

Cam Brown provided comments, feedback and

Parts of this thesis are presented in the following publications.

- A version of Chapter 2 is published:
 - Jaffray, R. (2025).
 - I was the main author on this publication. I did the following tasks. Dr. Gregory Paradis was the supervisory author who did these other tasks. Buddy also supported.
- A version of Chapter 3 is published:
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List of Abbreviations

SFM	Sustainable Forest Management
SFO	Sustainable Forest Operations
FOP	Forest Operational Planning
FRM	Forest Resources Management

List of Symbols

Σ summation

Acknowledgements

I am very grateful for...

Dedication

A dedication is a message from the author prefixed to a work in tribute to a person, group, or cause. Most dedications are short statements of tribute beginning with “To...” such as “To my family”.

Chapter 1

Introduction

1.1 Background and Significance

1.1.1 Sustainable Forest Management and Operations

Ecosystems services are the benefits people obtain from ecosystems [Hassan et al., 2005]. Growing recognition of ecosystem services has re-framed natural resource management from a focus on single-resource optimization toward maintaining the capacity of ecosystems to deliver multiple, interrelated benefits (ecosystem services) across spatial and temporal scales [Baskent et al., 2020]. In forest ecosystems, sustainable forest management (SFM) refers to the stewardship of forest ecosystems to maintain ecosystem services (e.g. timber production, carbon storage, biodiversity, water regulation, recreation and aesthetics) while balancing social, economic and ecological objectives [Thompson et al., 2021]. Researchers and practitioners alike have been vocal in their calls for change in sustainable forest management, initiating paradigm shifts for proactive management across multiple disciplines to encourage the inclusion of objectives like wildfire coexistence [Daniels et al., 2024], hydrology and watershed management [Green and Alila, 2012], climate change [Schoene and Bernier, 2012] and social license [Moffat et al., 2016] in SFM. Many of these shifting approaches to SFO have been documented in Polinko and Coupland [2021]’s bibliometric analysis of paradigm shifts in forestry.

These objectives are typically expressed at broad spatial and temporal scales, but the realization of these objectives depends on how forests are managed through day-to-day activities, referred to as forest operations [Weintraub and Romero, 2006]. Sustainable forest operations (SFO) are the operational implementation of SFM and are necessary to carry out larger time- and spatial-scale goals to be carried out sustainably while also contributing to sustainability themselves [Schweier et al., 2019]. SFO can be defined in Marchi et al. [2018]’s framework that includes technologies, methods,

systems and practices used to plan, carry out, monitor and improve these day-to-day activities, while accounting for five key performance areas: (i) environment; (ii) ergonomics; (iii) economics; (iv) quality optimization; (v) people and the society [Marchi et al., 2018].

Hierarchical Forest Management Levels

To translate forest management objectives into actionable decisions across scales that consider uncertainty and dynamic conditions, forest management utilizes a hierarchical planning system [Weintraub and Cholak, 1991]. This system separates decision-making across strategic, tactical and operational levels, which differ by spatial and temporal extents and levels of detail 1.1.

Strategic forest plans operate at the largest spatial and temporal scale, focusing on long-term management of objectives across the landscape that can span multiple centuries; in contrast, tactical plans operate at finer spatial and temporal resolution, include spatially linked information and span between five and twenty years [Belavenutti et al., 2019]. In terms of detail, tactical plans identify general areas and actions needed for forest management [Bettinger and Chung, 2004], but specific forest operations implementation details are identified in operational plans. At times, tactical and operational planning levels and specified planning horizons may overlap, particularly for smaller forest tenure holders or teams with limited planning capacity. For example, a small forest company may prepare a six-month harvest and access plan that they would refer to as their 'tactical' plan and a one-month harvest and access plan that would in greater detail outline the exact timing of their 'operational' plan. In such cases, the distinction between tactical and operational planning is defined by the function and level of detail of decisions being made rather than the absolute time horizon. Ba et al. [2016] asserted that if the order of tasks and their start or end time needs to be defined, then it is an operational decision. Using Ba et al. [2016]'s definition, operational planning focuses on the spatially-explicit allocation of resources (e.g. machines, workers, harvest systems) to achieve the goals outlined in the strategic and tactical plans [Church, 2007], including deciding exactly when and where to send harvesting equipment. Planning horizons at the operational level may vary depending on the structure of the model and the needs of the user. Operational plans are often associated with the shortest-term horizons in hierarchical plans; these plans can range from a daily horizon up to multi-month horizons [Epstein et al., 2007].

1.1. Background and Significance

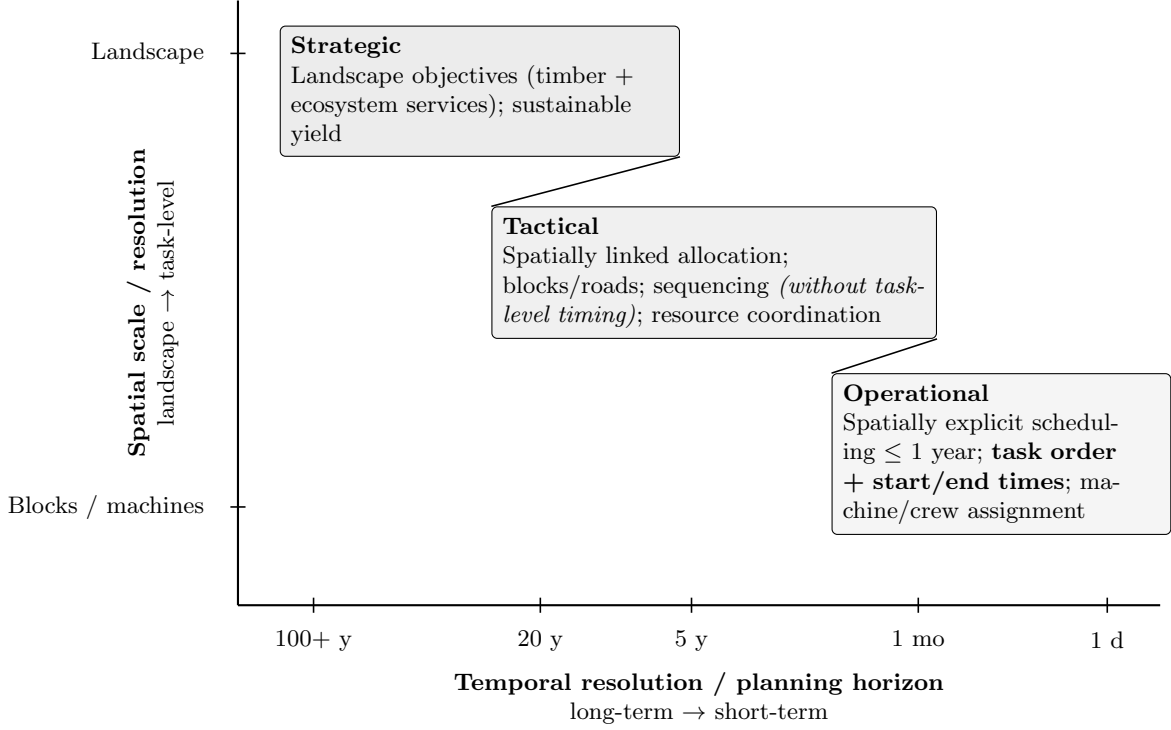


Figure 1.1: Hierarchical forest management planning levels shown by spatio-temporal resolution and decisions relevant to this research.

Uncertainty in Forest Management Planning

As emphasized by Röser [2012], forest management and forest operations take place within an operational environment shaped by social, environmental and economic conditions as well as uncertainty. Uncertainty and complexity are present across sustainable forest management and need to be considered in decision-making and their associated planning levels [Yousefpour et al., 2012].

At the strategic level, climate change introduces uncertainty around future growth and yield [McDowell et al., 2020], abiotic and biotic disturbances [Seidl et al., 2017] and long-term species viability [Clark et al., 2011]. Evolving societal expectations regarding the balance of ecosystem services as well as policy changes can shift management priorities and practices [Benner and Lertzman, 2022]. Market fluctuations can also affect long-term economic

viability of different products [Sohngen et al., 2019]. To deal with this uncertainty, scenario planning adaptive management approaches are used to plan for variance in future conditions [Mowrer, 2000].

At the tactical planning level, similar factors can also cause challenges. Abiotic and biotic disturbances can disrupt planned harvest sequences and road construction over the 5-20 year tactical horizon [Harvey et al., 2003]. Market changes can also affect what stands are economically viable to harvest and what resources (workforce, equipment, contractors) are available [Prestemon and Wear, 2000], which can also affect operational planning.

At the operational level, there are many sources of uncertainty. Weather conditions can quickly shut down operations, equipment can break down, ground conditions or access may turn out to be different than what was planned for and mill demands shift in real-time to influence what gets harvested. Environmental compliance can also induce immediate operational changes; for example, a wildlife observation could require buffering a nest out from the harvest area. Effects of these uncertainties can compound quickly at the operational level because of the compressed time frames while also working to ensure tactical and strategic goal completion. In forest operations, this has included adopting more adaptive forest operations strategies [Tampekis et al., 2024] and building stronger links between operational-level planning and longer-term planning levels [Bogle and van Kooten, 2015, Gautam et al., 2017].

Tactical and operational plans commonly have to be updated frequently as conditions change (weather, markets, breakdowns, etc) while staying aligned with long-term strategic goals. As a result, operational planning must remain adaptive in order to both achieve sustainability goals and maintain the feasibility of higher-level plans. With evolving sustainability expectations placing increasing demands on forest planning models, particularly at the operational level where Marchi et al. [2018]’s five performance areas must be addressed, models need to account for more details than just timber production. To support decision-making under these uncertain conditions, forest management planning uses various modelling approaches, including Operations Research methods.

1.1.2 Operations Research in Forestry and Forest Operational Planning Tools

Operations Research (OR) is the application of analytical and mathematical methods to support and improve decision-making [Hillier, 2005]. In forestry, OR methods provide a formal framework for representing objectives, constraints and trade-offs across planning levels and spatio-temporal scales. Methods such as mixed-integer linear programming models (MIPs), heuristics, meta-heuristics and simulation are used to address complex planning problems characterized by multiple objectives, limited resources, strict operational constraints and substantial uncertainty [Rönnqvist et al., 2015]. These methods help planners and forest professionals formalize objectives, explore solution spaces, quantify trade-offs, and create repeatable decision-support workflows.

The selection of an OR method depends on both the scale and structure of the problem as well as the objectives and decision context in which it is applied, including across hierarchical forest management planning levels. MIPs are often used to solve smaller problems with consistent structure, where exact solutions can be obtained within available computational capacity. Heuristic or meta-heuristic solvers are generally better suited to larger, more complex problems where exact solutions would be mathematically intractable or computationally infeasible [Bettinger et al., 2016]. In addition to these considerations, classical linear and mixed-integer programming relies on the assumption of certainty, where parameters such as cost, productivity and constraints are treated as known and fixed at the time of decision-making [Dantzig, 1963]. In forest operations, however, this uncertainty is rarely attainable. Solutions that are mathematically optimal under deterministic assumptions may perform poorly when implemented in dynamic and uncertain operational environments [Bertsimas and Sim, 2004].

To address these limitations, simulation-based approaches can be used in decision-support to complement OR optimization. Simulation can be used to test the robustness of plans and schedules under uncertainty, comparing alternative harvesting systems, and assessing sensitivity to changes in operational environments. By explicitly recognizing variability and system interactions, simulation can allow planners to evaluate how plans perform under a range of uncertain and dynamic conditions to provide insight into how the plans are likely to perform if implemented [Mowrer, 2000].

OR methods are applied across planning levels in forestry. At the strategic

level, OR supports long-term decisions such as determining sustainable harvest levels over time, rotation ages, and different silviculture systems and stand development pathways. At the tactical level, OR methods help allocate harvest blocks and coordinate resources (workforce, equipment, contractors) within the medium-term planning horizon. OR is also heavily used in supply chain planning in the flow of fibre from the forest onwards [Bouchard et al., 2017]. However, there is a lack of operational decision level models in forestry supply chain planning [Santos et al., 2019]. Operationally, applications of OR in forest harvesting include what areas will be harvested, what machinery will be used for harvesting and how the logs will be transported and handled [Epstein et al., 2007]. The application of OR in log logistics and transportation is a sufficiently complex and large-scale problem that it is frequently treated as a distinct subfield of OR applications in forestry [Epstein et al., 2007, Rönnqvist et al., 2015, Malladi and Sowlati, 2017]. However, given the scale and complexity of integrating harvesting and transportation, this research focuses on the decisions of machine scheduling and movement within and between harvest blocks, though it is recognized that landing placement, block layout and machine movement interact and influence transportation and log logistics decisions.

1.1.3 Significance of Forest Operational Planning Tools

Forest operational planning tools use OR methods to support short-term decision-making in forestry, including block harvesting and machine scheduling, crew allocation, processing and transportation planning. Together, these decisions are commonly known as the 'forest operational planning problem', a critical challenge throughout forest resources management and its modelling efforts [Martell et al., 1998]. Numerous solutions have been proposed to multiple sections of this problem, covering a range of decisions, operational environments, forest types and harvesting systems [Rönnqvist et al., 2023].

In this research, we tackle a small section of the forest operational planning problem: machine scheduling. Machine scheduling refers to the assignment of equipment to specific blocks, tasks and time periods in a way that is operationally feasible, aligns with longer-term plans, and is workable for operators and crews on the ground. These assignments are dependent on the operational environment and are shaped by user objectives, production targets and a range of regulatory, safety and environmental considerations and constraints. Operational planning tools provide the mechanism through

which these considerations are integrated and translated into actionable operational plans. Because these operational plans describe how and when harvesting activities occur on the ground, the quality of operational planning can directly influence whether forest operations contribute to SFM. Sustainable forest operations are therefore a necessary condition for achieving broader forest management objectives. Timber procurement may not be the primary objective emphasized by the public [Harshaw et al., 2009]; however, rising global wood consumption continues to require harvested trees [Arendarczyk et al., 2025]. Harvest operations can also support a broader range of objectives through active management of forests, including fuel reduction [Stephens et al., 2009], and innovative silviculture practices [D’Amato and Palik, 2021] among other ecosystem services [Sing et al., 2018].

The impact of operational planning decisions are often long-lasting on the landscape, in part because forest management operates over extended time horizons. Rotation lengths and the time required to move through successive planning stages mean that the outcomes of operational decisions may not be fully realized until decades later, when stands are next considered for harvest. Differences in operational planning quality can be observed in forest conditions that reflect decisions made years earlier under different management objectives and regulatory contexts.

In the shorter term, operational decisions directly influence the quality, volume and timing of logs delivered to mills, affecting mill profitability and product mix [Shahi et al., 2018]. These same operational decisions that shape mill supply in the short term can also influence ecological conditions that determine future forest productivity. Unsustainable practices have the ability to exacerbate reductions in future wood supply and impact soil and stand structure such that future stand composition, productivity, and resilience is compromised [Kneeshaw et al., 2000]. At the same time, these decisions can also affect forest regeneration and long-term stand productivity through effects such as soil compaction, which restricts root growth and limits future stand development [Worrell and Hampson, 1997]. Operational decisions in forestry have downstream effects on the immediate wood flows and product quality but also on the timing, volume and characteristics of future harvest opportunities.

The operational planning problem is complex with many interrelated outcomes to manage for. Operational planners are required to simultaneously consider short-term production targets, constraints, impacts and goals across spatial and temporal scales with high uncertainty and pressure. These

considerations must be addressed under tight planning timelines, making it difficult to explore and compare alternative courses of action using manual approaches alone.

Planning tools can support users by relieving some of the burden by structuring complex information and enabling faster comparison of alternatives. In this way, forest operational planning tools extend the ability of the practitioner, particularly when managing operations at the landscape-level with diverse stand conditions and evolving management priorities. These tools are important because they enable practitioners to make timely, defensible and sustainable decisions in an operational environment characterized by uncertainty and complexity. By supporting human experts in systematically analyzing and evaluating alternatives, these tools help support sustainable forest operations and their alignment with sustainable forest management.

1.1.4 Challenges and Limitations of Current Tools

Operations research was developed out of a growing need to increase efficiency in the planning of multi-part industry processes [Hillier, 2005]. In forestry, the diversity of operational environments, logistical constraints, harvest systems and management objectives has resulted in a wealth of literature [Pasalodos-Tato et al., 2013, Ezquerro et al., 2016, Weintraub and Romero, 2006] and wide range of proprietary decision-support tools designed to improve forest operational planning (FOP). Despite substantial advances in knowledge generation, both academic and industrial contributions are limited in their ability to be reused, extended or transferred between operations. For example, Henkelman [1978] developed a simulation model of timber harvesting operations in British Columbia (BC) whose representation of harvesting operations are still relevant 48 years after development. However, the knowledge included in this model is largely locked in its original implementation, which is not readily accessible, reusable or interoperable with modern data sources or software environments, requiring substantial re-engineering for use, effectively forcing researchers or users to reconstruct rather than reuse the original work.

During this time period, forest operations have undergone dramatic changes in data capture technology (e.g. GNSS, LiDAR, machine telematics), computing power and the formalization of forest operations as a distinct discipline [Heinimann, 2007]. Many legacy models were developed under assumptions of limited data availability and timber-centred management objectives, making them poorly suited to incorporate modern operational data. This

gap has been further widened with technological advancements, as modelling frameworks have not fully adapted to exploit the increasingly detailed operational data available.

Open-source tools have the potential to mitigate these limitations by improving transparency and reproducibility, but adoption and creation of open-source tools remains low. A major reason for the low uptake of open-sourced tools in FOP is likely the competitive advantage that proprietary software can provide [West, 2003]. Proprietary software often includes formal technical support as part of a commercial business model whereas many open-source projects operate without sustained financial backing for long-term support [Alami et al., 2024]. Open-source software often relies on voluntary contributions from maintainers who care about the project and its users [Mohit, 2025]. Thus, support for open-source software can be highly effective but is less predictable, as it depends more on personal motivation than guaranteed funding. Additionally, the more user-driven style of open-source tools as well as their higher programming literacy demands can reduce adoption or ease of use [Bravo et al., 2025]. With open-source software often relying on users to have high computer programming and coding literacy, this can add an additional barrier-to-entry in forestry where operational planning does not require a high level of programming [Holzinger et al., 2024]. Additionally, operational data is often has 'messy' structures that may not work well with a requirement of structurally consistent data input formats [Castellanos et al., 2017].

1.1.5 Forest Operations in British Columbia

British Columbia (BC) provides an ideal case study for the development and evaluation of forest operational planning tools because of its diversity and variability within the operational environment across ecological, economic, regulatory and social dimensions. As the most ecologically diverse province in Canada, BC has a wide range of site and stand conditions, operational constraints, access and disturbance regimes that all inform forest operational decisions [Dearden, 1983].

This ecological diversity is reflected in the heterogeneity of harvesting systems employed across the province. Ground-based, cable and helicopter systems are all used actively in British Columbia, sometimes within relatively close geographic areas, increasing the complexity of operational planning (Table 1.1). Ground-based systems cover the broadest range of BC's operating conditions and are the most cost effective. As such, ground-based

systems are the most widely used and have been able to expand into steep terrain with advancements in tethered-assist technologies and tilting equipment. Cable systems that suspend logs for primary transportation are used in steep terrain where sensitive soils are a concern (to minimize machine movement in the block) [Visser and Harrill, 2017] whereas helicopter logging is used selectively for high-value timber or where access is constrained [Grigolato et al., 2016]. This heterogeneity in harvest systems amplifies the need for generalized and adaptable planning tools. If a tool can accommodate the range of harvesting systems and terrain conditions within BC, it is likely that it would be able to adapted and generalizable to other jurisdictions.

BC has the largest forest sector in Canada, meaning that forest operational planning decisions can carry substantial economic consequences. In 2023, for example, BC’s forest sector generated 50,225 jobs and over \$5 billion in GDP [Ministry of Forests, Economics and Trade Branch, 2023]. As a primarily commodity-based sector, forestry contributes to 20% of BC’s commodity exports [Ministry of Forests, Economics and Trade Branch, 2023]. This economic importance is coupled with exposure to market and supply variability [West et al., 2021], which directly influences the feasibility and timing of forest operational planning decisions. These challenges are further exacerbated by abiotic and biotic disturbances including record-breaking wildfire seasons and the mountain pine beetle epidemic [Axelson et al., 2009]. This volatility has highlighted the importance of efficient and reliable SFO that are aligned with higher-level plans while accounting for ecological diversity and economic risk [Bańkowski, 2025].

A defining characteristic of forestry in BC is change. Henkelman [1978] proposed the model as a means for examining change with the at-the-time current structure of forest operations in Gold River, BC. Change continues to be a constant in BC forestry [Bonita, 1972], but the drivers of change are more pronounced. Climate change impacts, shifting regulatory frameworks and increasing disturbance frequency have expanded the need to incorporate uncertainty across hierarchical planning levels. Forest managers in BC increasingly face decisions under high-risk conditions, requiring tools capable of evaluating operational strategies across multiple possible futures and pathways [Day and Pérez, 2013].

Regulatory and governance complexity further contributes to uncertainty in SFO. Approximately 95% of the timber in BC is on public land [Government of British Columbia, Ministry of Forests, 2025], meaning that forest oper-

ations are strongly influenced by public policy and evolving public values. As public interest shapes policy, regulatory changes can significantly affect operational decision-making. Forest operations also support many forestry-dependent communities and can have a direct impact on local employment and economic stability. Social license, particularly in a jurisdiction where most managed forests are publicly owned, has become increasingly important, alongside the growing recognition of Indigenous rights and role in forest governance.

Current legislation and policy frameworks establish expectations for operational planning to be aligned with higher-level objectives, such as the requirement for Forest Operations Plans to conform with the Forest Range and Practices Act (FRPA) [Government of British Columbia, Ministry of Forests, 2025]. Additionally, the practice of professional forestry in BC is regulated and requires registered professional foresters to uphold public interest, which without an explicit definition has adds flexibility and significant responsibility onto foresters to plan and operate transparently and defensibly [Robson, 2024]. These regulatory conditions apply across both public and private land, increasing the value of transparent and generalizable planning tools that can be applied consistently across contexts.

Finally, tenure diversity within BC’s forests adds complexity to operational planning. In addition to the differentiation between public and private land in the forest sector, there is also tenure diversity within public land operations. Tenure diversity (a mix of players on the landscape from major licensees, community forests, First Nations Woodlands Licenses and small-scale operations) creates substantial variation in management capacity and risk tolerance in decision-making. Large tenure holders often have the flexibility to shift activities across broader areas, while smaller community-based operations often experience tighter spatial, financial and operational constraints [Devisscher et al., 2021]. An effective forest operational planning tool must accommodate these variations in organizational scale and decision-making contexts.

BC was chosen as a valuable and representative case study for developing and evaluating adaptable and generalizable forest operational planning tools. There are many opportunities within the province where planning efforts need to respond to shifting conditions, work within constrained operational environments and accommodate a variety of needs. The diversity of operations in BC means that scheduling and operational planning needs differ. Should a model be presented that can function within this environment,

it would likely be applicable elsewhere.

1.2 Motivation

1.2.1 For a Systematic Review

There have been many models, tools and decision-support systems (DSSs) developed to support in forest operational planning. The existing reviews covered different scopes within forest operations and transportation planning but to our knowledge, no review existed that covered the strengths, limitations and barriers to adoption of forest operational planning tools. Therefore, the need for a structured review of forest operational planning tools is present.

A systematic review was selected to conduct the review of literature to ensure transparency and reproducibility in the identification and evaluation of planning tools, similar to the overarching goal of supporting transparent and reproducible tools in forest operational planning. Bibliometric analyses would not be suitable for this purpose as they assess the metadata and citation metrics of the publications instead of the technical modelling details that are needed to inform future model development. The application of systematic search methods also supports the inclusion of underrepresented models that may not be found using more ad hoc or narrative approaches. Models that have limited visibility may contain functional characteristics that prove useful in forest operational planning. A systematic review that includes both widely adopted and lesser-known models can help identify the strengths, limitations and areas for improvement across the solution space to inform future model development.

1.2.2 For an Open-Source Model

Decision-support systems (DSSs) are intended to support decision-making rather than replace those who are making the decisions [Seely et al., 2004]. In forest operational planning, DSSs are useful to support navigating complex and uncertain operational conditions. As forest operations move towards digitalization within the frameworks of Industry 4.0 and 5.0, there is further integration of human-centred design and ergonomic design in modelling [Marchi et al., 2018, Röser, 2012, Bolis et al., 2025]. For DSSs to be effective at the operational level, they must be usable, accessible and aligned with the realities of operational planning and execution.

Table 1.1: Description of Common Harvesting Systems in BC

Terrain/System	Felling	Primary transport	Processing	Notes
Ground-based	Feller-buncher	Grapple skidder	Roadside processor (stroke/dangle)	Loader → Trucks
Ground-based	Chainsaw (hand fall)	Shovel logger (hoe-chucker)	Roadside processor	Loader → Trucks
Ground-based	Feller-buncher	Shovel logger (hoe-chucker)	Roadside processor	Loader → Trucks
Cut-to-Length (CTL)*	Harvester (single-grip)	Forwarder	Harvester (single-grip)	Direct to Trucks (shortwood)
Steep-slope mechanized (tethered)	Winch-assist harvester or feller-buncher	Tethered shovel / tethered skidder / forwarder	Roadside processor	Loader → Trucks; includes anchor/winch unit
Cable – standing skyline (carriage + chokers)	Hand/mech fall	Skyline yarder (carriage + chokers)	Landing processor / Hand buck	Loader → Trucks
Cable – running skyline (grapple)	Hand/mech fall	Grapple yarder (swing/tower; motorized grapple carriage)	Landing processor / Hand buck	Loader → Trucks
Helicopter logging*	Hand fallers	Helicopter (longline)	Hand falling	Possible direct to water for transport, otherwise to landing

Note: Rows marked with * follow *felling/processing* then primary transport. All others follow *felling*, primary transport, then processing.

1.2. Motivation

The evaluation of current tools and their transparency and adaptability will support the development of a new modelling framework. These constraints restrict organizations’ ability to understand how decisions are generated. An open-source modelling framework presents a unique opportunity — beyond cost savings — to improve transparency and reproducibility; by providing an accessible framework developed for re-usability, organizations can track decision-making processes over time, learning what works and what doesn’t in different scenarios [Dahlander et al., 2021]. This continuity is valuable in uncertain contexts like forest operations, where it is helpful to apply recourse and revise decisions as conditions change in forest operational planning [Myers, 2004]. This is particularly important in hierarchical forest planning, where these plausible emulations of operational planning can be inputted into higher-level planning tools to anticipate operational level decisions at the strategic and tactical levels [Nelson et al., 1991].

Considering the rapid advances in artificial intelligence and computational optimization, many forest planning problems that were previously intractable can now be solved more efficiently. As computational constraints diminish, the bottlenecks in SFM are more associated with the translation of management objectives, values and constraints into model structures and parameters [Radke et al., 2020]. This translation process cannot be automated as it depends on critical thinking, domain expertise and relationship-building among planners, stakeholders and rights-holders [Nocentini et al., 2017].

Forest planning models can provide decision-support tools capable of quickly computing and evaluating complex scenarios, while simultaneously freeing time and cognitive capacity for participatory planning, community engagement and deliberative processes that cannot be automated [Devisscher et al., 2021]. Advances in modelling and computation can actually be used as enablers of more transparent and socially grounded forest management by freeing planners to focus on aspects of decision-making that cannot be delegated to automated systems [Tampekis et al., 2024].

Within the scope of this research, the goal is to present a proof-of-concept model for machine scheduling in forest operational planning. The prototype can demonstrate the feasibility of using open-source tools to support operational decision-making while aligning with the broader goals of transparency and human-centred forest management. Together, these motivations directly inform the structure of this research: first, by systematically assessing the state of current tools and second, by exploring the potential of open-source modelling approaches for machine scheduling in forest operational planning.

1.3 Research Questions and Objectives

This research investigates the potential of using an open-source modelling framework to support forest operational planning at a machine-scheduling level in forest operations in British Columbia, with the overall aim of advancing transparency and adaptability in forest operational planning.

1.3.1 Research Questions

The main question guiding this research is: Is there a generalized formulation of the forest operational planning problem that can be adapted to multiple operational environments in British Columbia?

Secondary objectives and questions are to explore the applicability of this forest operational planning model framework as it anticipates and responds to future challenges, approaching this question: How can simulation modelling frameworks improve agility of forest operational planning in anticipation of future challenges? To answer these questions, we have defined two research objectives.

1.3.2 Objective 1: Chapter Two

The first objective of this research project is to conduct a systematic review of operations research-based solutions in forest operational planning. This objective addresses the primary research question by identifying the design requirements and informing the design choices for a generalized open-source model in forest operational planning.

1.3.3 Objective 2: Chapter Three

The second objective of this research project is to design and evaluate the applicability of a generalized, open-source model for machine scheduling in forest operational planning. This objective addresses the primary research question by testing generalizability across operational contexts and testing proof-of-concept of using these tools for forest operational planning. This objective addresses the secondary research question by assessing how simulation-based frameworks can improve agility under future uncertainty.

1.4 Thesis Structure

The introduction outlines the importance of forest operational planning in achieving sustainable forest management and lays the framework and motivation for the work presented in Chapters 2 and 3.

Chapter 2 provides a systematic review of existing forest operational planning tools and models. Chapter 2 is formatted as a scientific article and has been submitted for publication in a peer-reviewed journal. This review presents the current state of forest operational planning tools with specific reference to their adaptability and re-usability and acknowledges gaps in current practice that directly inform the design choices of the open-source model presented in Chapter 3.

Chapter 3 presents the open-source and generalized tool for machine scheduling in forest operational planning and the applicability of simulation modelling in improving agility. The Forest Harvest Operations Planning System (FHOPS), is an open-source, generalized machine-scheduling proof-of-concept framework that treats site-specificity as an input rather than a design limitation. Using BC-style operational scenarios, we evaluate rolling-horizon planning to examine how far ahead planners need to schedule and how frequently plans should be updated. FHOPS provides a transparent way for planners to test “what-if” scenarios and compare options before committing operationally. In this chapter, the objective is to provide evidence-based guidance for practitioners on planning window selection and re-optimization strategies when applying FHOPS within the BC forestry context by evaluating rolling-horizon performance bounds with semi-synthetic datasets. This chapter is formatted as a scientific article and will be submitted for publication in a peer-reviewed journal.

Chapter 4 concludes this research project and proposes areas of future research and application for the proposed tool as well as more broadly the adoption of open-source tools in forest operational planning.

Chapter 2

Systematic Review

This chapter presents an article named *Forest harvesting operational planning tools: a systematic review of optimization, simulation, and spatial decision support systems*, submitted for publication to the International Journal of Forest Engineering. The authors are Rosalia Jaffray, Kathleen Coupland and Gregory Paradis.

This chapter is written as part of a thesis-by-paper approach and provides a systematic review that informs the modelling approach developed in Chapter 3. The general background and motivation for this work are outlined in the thesis introduction with this chapter focusing on a detailed assessment of existing modelling tools in forest operational planning. Please see the Overleaf titled *Rosalia Jaffray MASc Review Paper* for this chapter.

2.1 Thesis-Specific Additions

2.1.1 Linkages to the BC Context

While many studies are affiliated with Canada either through authorship or study site, there are also other tangible connections that can support the application of other models in the Canadian and more specifically, the British Columbia (BC) context. Viana et al. [2023] presents scenario modelling of different harvesting 'fronts', focusing on the interactions between contractors across a large land-base. Parallels can be drawn between this case study and Timber Supply Areas in British Columbia where we have multiple contractors and players who may operate with different skillsets and operating areas within the landbase. Other Canadian contexts like the Alberta Forest Management Area (FMA) system or Irving in New Brunswick would also have similar applications.

Chapter 3

An Open-Source and Generalized Model for Machine Scheduling in Forest Operational Planning

This chapter presents an article named *NAME OF ARTICLE*, submitted for publication to the JOURNAL NAME. The authors are Rosalia Jaffray, Kathleen Coupland and Gregory Paradis.

This chapter is written as part of a thesis-by-article. Please see the Overleaf titled *Rosalia Jaffray MASc Modelling Paper* for this chapter.

Chapter 4

Conclusion

This concluding chapter summarizes the contributions of this thesis, presents limitations and areas of further research as well as providing final remarks.

4.1 Summary of Findings and Implications for Practice

The first objective of this project was to test the proof-of-concept of an open-source reusable modelling framework to address and improve accessibility and reproducibility of forest operations modelling, to better support future work of researchers and practitioners in response to the question: Is there a generalized formulation of the forest operational planning problem that can be adapted to a series of operational environments in British Columbia? This objective was addressed in chapter 2 and 3, first systematically reviewing the current tools available and then presenting and evaluating a tool for forest operational planning. The secondary objective was to explore the applicability of this forest operational planning model framework as it anticipates and responds to future challenges, approaching the question: How can simulation modelling frameworks improve agility of forest operational planning in anticipation of future challenges? This objective was addressed in Chapter 3, applying the tool developed to a series of case studies and solving using multiple solution techniques to evaluate the tool's performance.

4.1.1 Chapter Two: Systematic Review of Operations Research-based Solutions to Forest Operations Planning

Chapter Two established foundational knowledge in forest operational planning tools by systemically reviewing the available literature. Using a systematic review approach, we identified the strengths, limitations and barriers to adoption of forest operations planning models. Of the models and tools we

reviewed, site-specificity was maintained as a key component of forest operational planning models. However, it was recognized as a limiting factor in using these models for any other site besides the one they were specifically developed for. The structural design approach of the models largely employed a hard-coding approach rather than including site-specificity as a design parameter. In addition to this common design decision, there was a recognized limited application of open-source and accessible tools despite a recognized need. Following the review of the literature, a design checklist was developed to inform future model development in an effort to support more accessible and usable tools for forest practitioners.

4.1.2 Chapter Three: An Open-Source Generalized Framework for Forest Operations Planning

Building off of the strengths, limitations and design checklist presented in Chapter 2 in the current state of forest operational planning models, Chapter 3 presents and evaluates an open-source generalized modelling engine for machine scheduling in forest operational planning.

- What did the modelling engine represent
- What was evaluated
- What the evaluation showed
- Implications

Focusing on the development and formulation of a generalized engine for exploring the forest operational planning solution space,

The modelling framework can produce at greater efficiency a range of possibilities that forest managers can choose from, connecting technological development to the needs of the operational planning forester.

- Discussion will be SWOT focused, including highlighting areas of opportunity to use these mapped trade-offs to communicate forest management decisions out to the public as well as add to the discussion of what 'good' operational plan options look like from an OR lens.
- This framework can be used to test 'what-if' scenarios of the counterfactual, allowing forest managers greater flexibility in assessing the impact of innovative ideas (e.g., changes to equipment, different scheduling, impact of fires) on the operational planning. These contributions (tools) can inform decision-making, allowing current operational

planning strategies to be modelled and compared with the outputs of different management alternatives.

- The development of an open-source modelling framework contributes to accessible forestry as related to transparency, social license conversation etc Value of accessibility as a form of extension, allowing for more transparent information sharing of forestry practices. I would argue that an open-source model in forest operational planning is a form of anticipatory extension in forestry as it takes steps towards making these planning efforts accessible and supports information sharing and decision-making for forest practitioners [Hoffman et al., 2025] accessibility and diversity of materials that support information sharing, learning, and decision-making for forest practitioners.
- Can be used by grad students and other academic uses to explore operational planning including using it to create outputs that can be loaded into strategic planning tools to further bridge between strategic and operational planning levels, strengthening those available linkages

4.1.3 Integrated Findings

- We identified systemic design barriers in forest operational planning tools in Chapter Two and then in Chapter Three, demonstrated a proof-of-concept for a solution to solve or attempt to solve some of these challenges and evaluated this proof-of-concept tool's rolling-horizon performance limits.
- Showing that the generalized formulation is feasible and adaptable to a variety of operational environments, using BC as the case study with the acknowledgement that if it can work in BC it should be able to work elsewhere.
- Showing how open-source planning tools can support agility, transparency, and knowledge transfer
- Implications for future research are denoted in section 4.2.

4.1.4 Limitations

- Can use RESULTS in the future and validate findings using past operational plans that have been harvested

4.1.5 Practical Implications

- Linkages of cost and productivity for more integrated planning
- Scenario testing, trade-off mapping, using by practitioners
- Alignment with provincial initiatives for accessible decision support tools
- How open-source tools support transparency, communications, public trust

4.2 Further Research

The development of an open-source and generalized tool for machine scheduling in forest operational planning represents an important step towards improving the transparency and adaptability of forest operations. This thesis demonstrates the proof-of-concept of this tool and this section outlines several opportunities for future research and tool development to strengthen its contribution to sustainable forest operations and management. Additionally, in each chapter, more specific opportunities for future research are linked to their respective articles.

4.2.1 Scenario Analysis for Operational Innovation

Using the model to enable accurate scenario modelling and analysis of the counterfactual without incurring implementation costs. This is nested within a broader appreciation that the computational approaches will allow us to explore what we haven't before using this proposed combination of computational and quantitative measures. This framework can be used to test 'what-if' scenarios of the counterfactual, allowing forest managers greater flexibility in assessing the impact of innovative ideas (e.g., changes to equipment, different scheduling, impact of fires) on the operational planning. As echoed in Boston and Bettinger [2001]'s recognition that the increased change in regulatory and political drivers of change alongside climate change can result in more companies asking "what-if" questions. This can be a useful tool to ask these questions and to explore these decision spaces quite quickly. By exploring the potential effects of these decisions using the model, more testable and defensible guesses. These contributions (tools) can inform decision-making, allowing current operational planning strategies to be modelled and compared with the outputs of different scenarios.

- Extensions for what-if scenarios (alternative machine or system mixes, altered work shifts) to explore operational environment implications and trade-offs linked to occupational, mechanical, ecological, economic trade-offs.
- Nested within the broader appreciation that the computational approaches will allow us to explore what we haven't before using this proposed combination of computational and quantitative measures.
- Exploring switching operational equipment over from buncher/skidder to harvester/forwarder. What are the cost analyses of different contractors changing their equipment roster over from one harvesting system to the next? How does this improve or not improve costs?
- Using this as a tool for exploring this decision tree as part of scenario analysis for innovative practices.
- Opportunity for testing different operational guidelines and best management practices prior to field implementation to better inform policy development. Making it more adaptive.
- Can be used by a Canfor for scheduling of multiple operators and logging contractors or can be used by a single operator looking to test out and simulate operational planning decisions and changes without actually investing the capital required to make those changes

This is nested within a broader appreciation that the computational approaches will allow us to explore what we haven't before using this proposed combination of computational and quantitative measures. Exploring switching operational equipment over from buncher/skidder to harvester/forwarder. What are the cost analyses of different contractors changing their equipment roster over from one harvesting system to the next? How does this improve or not improve costs? Using this as a tool for exploring this decision tree as part of scenario analysis for innovative practices. Opportunity for testing different operational guidelines and best management practices prior to field implementation to better inform policy development. Making it more adaptive. Sensitivity modelling.

4.2.2 Multi-Objective Optimization

- Incorporating multiple objectives into the machine scheduling tool could allow for different objectives to be solved and their solutions to be compared so that people can see how these impact their final

machine schedules. For example, comparing cost efficiency vs reduced carbon emissions scenarios may inform more holistic forest operational planning objectives as compared to the dominant economic objectives presented currently.

- Pareto-optimal solutions
- trade-off analysis
- Expanding of the current productivity and cost integration components

4.2.3 Global Testing and Validation

- Additional testing and validation using multiple case studies in different operational environments ranging from native and plantation forests, different terrains, etc.
- Scalability across spatial environments.
- Expansion outside of the BC context

4.2.4 Stochastic Expansion

- Trying out a broader range of stochastic testing to see impact on operational decision-making for forest operational planning. Embedding more stochastic elements into machine scheduling so that there is more buffer against uncertainty
- Further integration of stochastic and probabilistic modelling techniques to explicitly represent operational uncertainty
- Initial inclusions of Monte Carlo simulations. Expanding stochastic modelling approaches used
- The smaller the operator, the more exposed they are to statistically fuzzy inventory data that they use in operational planning processes. The ability to quickly explore feasible alternatives (recourse) is critical.

4.2.5 Graphical User Interface

- Recognized programming literacy gap within forestry and forest operational planning
- GUI could improve uptake and usage to go beyond academic scopes

4.2.6 Linkages within Operational Planning

- Connecting with log logistics and trucking planning or secondary transport planning
- Complementary research with open-source trucking logistics models to test the impact of harvesting decisions on transportation and vice versa
- For example, how does the spatial distribution of harvesting impact transportation costing or route scheduling? Inversely, additional studies could explore the impact of different trucking and fleet configurations (e.g. percent electric, payload limitations) on optimal harvest scheduling design.

4.2.7 Linkages with Other Planning Levels

- Interoperability with other tools including strategic landscape forest estate modelling
- Scalability across temporal range
- Connecting operational and strategic decision-making levels, nesting decisions within the hierarchical forest management levels
- Beaudoin et al. [2008] for linkages between planning levels

4.2.8 Integration of Real-Time Operational Data

Productivity and cost are critical factors in operational planning and the expansion/integration of real-time operational data could have some great leads for this project to expand its operational feasibility.

- Productivity variability and accurate productivity values are currently static in the current model. The model may be better improved using advancements in real-time productivity tracking that can continuously update and refine model simulations.
- Integrating sensor data to adjust machine scheduling dynamically, responding to unplanned events like breakdowns or weather, gaining real-time data from telematics or IoT platforms
 - Similar to [Keefe et al., 2022] for connection to machine scheduling and real-time data, increased digitalization in forestry as a the future

4.3 Final Remarks

It is the objective of this thesis project to provide proof-of-concept that accessible and open source modelling frameworks have a place in forest operational planning. The findings presented here illustrate how such a framework can support sustainable forest operations and we hope that this will inspire continued adoption of these transparent and open practices.

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Appendix A

Sentences I May Want To Use Later

Across planning levels, forest management has shown a documented shift from timber-oriented decision-making to more holistic planning that explicitly incorporate a broader range of social, economic and environmental values [Weintraub and Bare, 1996].

Scheduling problems in forestry share similar structures with flow-shop and job-shop scheduling problems studied in other resource-extraction sectors like mining [Åstrand et al., 2018]. In addition, other sectors provide useful approaches for integrating uncertainty across operational and tactical levels [Song et al., 2015]. These parallels suggest opportunities to adapt and extend established scheduling frameworks to FOP, provided that they can accommodate the unique spatial and operational characteristics in forestry. Drawing on other sectors that also have dynamic operating conditions and high uncertainty, there are opportunities to apply these methods and approaches to forest operational planning. In forest operational planning, there are many sources of uncertainty.

Referring back to Ba et al. [2016]’s definition, operational decisions involve defining the order of activities as well as their start and end times. Depending on the harvest system used, there is a well-defined sequence of operations required to transform standing trees into logs that can be transported to processing facilities. This can be translated into models using precedence constraints that ensure downstream activities cannot occur before prerequisite operations are completed, as seen in Arora [2023], where the model includes constraints that make sure that volume cannot be primary transported as a log before a tree is felled to become a log. Additional timing decisions may include restrictions on when harvesting can begin in a block to avoid seasonal access restrictions or the coordination of machine deployment with the completion of road construction, as seen in citation.

In this research, we tackle the small section of the forest operational planning problem that includes machine scheduling. Machine scheduling refers to the assignment of equipment to specific blocks, tasks and time periods in a way that is operationally feasible, aligns with longer-term plans, and is workable for operators and crews on the ground. These assignments are dependent on the operational environment and are shaped by user objectives, production targets and a range of regulatory, safety and environmental considerations. Machine scheduling decisions are also subject to a variety of constraints including:

- Uncertainty
- Mobilization costs
- Production requirements (even-flow)
- Resource constraints
- Factors that affect machine type (terrain, piece size)
- Timing (wildfire scheduling, precedence, the five seasons)
- Not a constraint but I do mention the regulatory, safety and enviro constraints earlier without including them in the constraint list so I shouldn't do that prob

Over the past decades, machine scheduling models have evolved with increases in computing power and data capability, but at the core remain site-specific with little adaptability built in to apply the built models to be used with other similar objectives, constraints or in different operational environments. The increasing level of complexity paired with the above advancements in technology that this adaptability should be more accessible than ever.

A simple example of operational complexity that these planning tools need to manage for is the impact that road conditions and weather forecasts have on the day-to-day feasibility of a forest operation plan. Different models and tools account for uncertainty differently, whether through explicit recognition or post-processing inclusion if they are not included in the modelling. As operational environments become more and more complex, the need for sophisticated planning tools that can support decision-makers becomes increasingly critical.

Appendix B

Name of Second Appendix